

Horizon Quantum Holdings

Quantum Software Compiler + Circuit Optimisation — Singapore-Based Deep Tech

PhD Due Diligence Report | BLRI-DD-HQ-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Quantum Compiler Technology: The Invisible Infrastructure Layer

Horizon Quantum builds quantum circuit compilation and optimisation software — the 'translation layer' that converts high-level quantum algorithms into hardware-native gate sequences. This is infrastructure software that every quantum hardware company needs. In Seldon K-variable terms: compiler technology is an enabling multiplier — it makes ALL quantum hardware more efficient without the capital intensity of building hardware. Horizon's Singapore base positions it in the ASEAN technology corridor with proximity to regional government quantum programmes.

moomoo OpenD data [2026-08-13]: Price \$14.50 | Mkt Cap \$747.9M | EPS TTM -\$0.349 | Net Income -\$18.0M | 52w: \$8.29-\$45.00 | Prev Close \$14.93 | Shares 51.6M

Source: moomoo OpenD [2026-08-13] · Horizon Quantum investor relations · Singapore NRF

PART II — QUALITATIVE + QUANTITATIVE

Singapore Quantum Engineering Programme (QEP)

Horizon is a beneficiary of Singapore's National Research Foundation Quantum Engineering Programme — a S\$25M+ initiative to develop quantum software and hardware capability in Singapore. As a home-grown quantum software company, Horizon receives preferential access to NRF grants, A*STAR research

collaboration, and government procurement. The ASEAN quantum market is nascent but growing: Singapore's Smart Nation initiative prioritises quantum for financial services.

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]
Price: \$14.50 | Mkt Cap: \$747.9M | EPS TTM: -\$0.349 | Net Income: -\$18.0M
52w: \$8.29-\$45.00 | Prev Close: \$14.93 | Shares: 51.6M

Metric	Value	Source
Price	\$14.50	moomoo OpenD
Market Cap	\$747.9M	moomoo OpenD
EPS TTM	-\$0.349	moomoo OpenD
Net Income	-\$18.0M	moomoo OpenD
52-Week Range	\$8.29-\$45.00	moomoo OpenD
Shares	51.6M	moomoo OpenD

PART III — COMPETITIVE + ETHNOGRAPHIC

Competitor	Type	vs HQ
Quantinuum TKET	Open-source compiler	TKET has 500K+ users; HQ has hardware-agnostic edge
IBM Qiskit	Open-source	Dominant; HQ differentiates on optimisation depth
Classiq (private)	Quantum compiler	Direct competitor; HQ listed; Singapore government moat

FINDING F-5-A: Portrait 1: The Singapore MAS Quantum Pilot Team (COMPOSITE)
A technology officer at the Monetary Authority of Singapore evaluates quantum circuit optimisation for financial risk models. Horizon's Singapore base and MAS relationship gives it privileged access to Singapore's regulated financial sector quantum pilots. 'We trust local vendors for sensitive financial applications.' Represents Singapore's financial services market — a premium B2B channel.
COMPOSITE / ILLUSTRATIVE — drawn from Singapore fintech quantum patterns

FINDING F-5-B: Portrait 2: The ASEAN Government Quantum Officer (COMPOSITE)
A technology policy officer at Singapore's Infocomm Media Development Authority channels NRF quantum grants to Horizon's compiler research. 'Horizon is Singapore's home-grown quantum software champion. We want to be a regional hub.' Represents Singapore's strategic quantum investment that gives Horizon non-dilutive funding.
COMPOSITE / ILLUSTRATIVE — drawn from Singapore government quantum policy patterns

FINDING F-5-C: Portrait 3: The Quantum Hardware Developer (COMPOSITE)
A quantum hardware startup in Australia integrates Horizon's compiler to reduce gate count on their superconducting system by 25%. 'Horizon's compiler is hardware agnostic and genuinely reduces our circuit depth. It's the toolkit we needed.' Represents the pan-APAC quantum hardware developer community that needs compiler tooling.
COMPOSITE / ILLUSTRATIVE — drawn from Asia-Pacific quantum developer patterns

PART IV — ADVERSARIAL + SUPERFORECASTING

HH1: TKET/Qiskit open-source compiler commoditises compiler market

EVIDENCE FOR:

Quantinuum TKET and IBM Qiskit are open-source and improving rapidly; free tools reduce willingness to pay for proprietary compiler

EVIDENCE AGAINST:

Proprietary compiler depth > open-source for hardware-specific optimisation; enterprise SLA support is billable where open-source is not

QUANTITATIVE TEST

P(HQ loses >50% of enterprise compiler revenue to open-source by 2028)=35%

VERDICT: BEAR — open-source commoditisation is real

Implication: Differentiate on depth + support; Singapore government moat

HH2: Singapore government mandates Horizon for national quantum programme

EVIDENCE FOR:

NRF designates Horizon as national quantum software champion; all Singapore government quantum pilots must use Horizon compiler

EVIDENCE AGAINST:

Government mandate risk for foreign competitors; ensures revenue floor

QUANTITATIVE TEST

P(Singapore government mandate by 2027)=30%

VERDICT: BULL — government mandate is transformative for small company

Implication: Monitor NRF/IMDA quantum programme announcements

HH3: Classiq (private) outcompetes and captures enterprise market

EVIDENCE FOR:

Classiq has \$51M in VC funding; strong US enterprise traction; global team

EVIDENCE AGAINST:

HQ has Singapore government moat; ASEAN market specialisation; lower competition

QUANTITATIVE TEST

P(Classiq captures >30% of quantum compiler market including HQ's target segment)=25%

VERDICT: BEAR-NEUTRAL — Classiq is a direct threat in non-Singapore markets

Implication: Monitor Classiq enterprise wins in Asia

HH4: HH4: Acquisition by IonQ or Quantinuum for compiler IP

EVIDENCE FOR:

Both IonQ and Quantinuum need compiler technology; HQ's \$747.9M market cap is acquirable

EVIDENCE AGAINST:

Hardware companies may prefer open-source compiler integration to acquisition

QUANTITATIVE TEST

P(hardware quantum company acquires HQ by 2027 at >30% premium)=15%

VERDICT: BULL — acqui-hire of compiler IP would be premium deal

Implication: Monitor partnership announcements with hardware companies

HH5: HH5: Small float + Singapore listing creates extreme volatility

EVIDENCE FOR:

51.6M shares only; limited institutional coverage; sentiment-driven

EVIDENCE AGAINST:

Small float can amplify positive catalysts — government mandate would spike 50%+

QUANTITATIVE TEST

P(>40% price move on single news event within 12 months)=55%

VERDICT: NEUTRAL — size both risk and opportunity

Implication: Position sizing essential; options not liquid enough for hedging

SCENARIO 1: Singapore national quantum software mandate (P = 28%%)

NRF/IMDA designates Horizon for national quantum pilot; \$5M+ government contract; stock to \$22-25

Driver: NRF budget announcements

Falsifier: Other vendor selected; mandate delayed past 2027

SCENARIO 2: Enterprise compiler sales: 10+ paying customers (P = 25%%)

Horizon signs 10+ enterprise compiler licences (\$100K-500K/year each); ARR reaches \$2M+; first path to profitability visible

Driver: Customer announcements; ARR disclosures

Falsifier: Qiskit/TKET dominance; no paying customers by end 2026

SCENARIO 3: Base case: survival + NRF grants (P = 35%)

Horizon maintains NRF grants; compiler used in academic/government pilots; losses narrow to -\$10M; stock \$10-18

Driver: NRF grant renewals; pilot programme announcements

Falsifier: Grant non-renewal; revenue below \$1M

SCENARIO 4: ASEAN quantum hub strategy delivers B2B contracts (P = 15%)

Thailand, Indonesia, Malaysia governments select Horizon for national quantum pilots; ASEAN revenue stream opens

Driver: ASEAN government quantum programme announcements

Falsifier: ASEAN nations select IBM/IonQ for government quantum

SCENARIO 5: Acquisition at premium (\$20-25/share) (P = 12%)

IonQ or Quantinuum acquires Horizon compiler IP; \$20-25/share acquisition

Driver: Hardware company M&A; partnership expansion signals

Falsifier: No acquisition interest; companies build compiler in-house

SCENARIO 6: Quantum winter — stock retests \$8.29 low (P = 18%)

Broad quantum selloff; Horizon pre-revenue; 52w low retested

Driver: Two quarters of no material revenue

Falsifier: Singapore mandate announcement reverses selloff

SCENARIO 7: Financial services quantum pilot (MAS) (P = 20%)

MAS selects Horizon for Singapore Central Bank quantum computing pilot; credibility catalyst; stock +30%

Driver: MAS quantum initiative announcements

Falsifier: MAS selects IBM Quantum or IonQ for pilot

SCENARIO 8: Open-source TKET community adopts Horizon extensions (P = 15%%)

Horizon contributes hardware-specific optimisation plugins to TKET; gains 5,000+ developers; brand recognition builds

Driver: TKET GitHub contribution activity

Falsifier: Community rejects proprietary extensions; forks Horizon's work

SCENARIO 9: Series B funding from SoftBank/GIC at premium valuation (P = 10%%)

GIC (Singapore sovereign wealth) or SoftBank invests in Horizon at \$1B+ valuation; non-dilutive to public shareholders

Driver: GIC technology investment portfolio; SoftBank quantum strategy

Falsifier: Investors decline; Horizon dilutes at \$12-14

SCENARIO 10: Bull case: \$1B valuation as ASEAN quantum compiler standard (P = 4%%)

Horizon becomes standard quantum compiler across ASEAN government programmes; enterprise ARR \$20M; profitable path

Driver: All government + enterprise catalysts fire

Falsifier: Open-source dominates; Horizon cannot monetise compiler

PART V — SELDON + RISK REGISTER

Variable	HQ Impact	Direction	Magnitude
K	Compiler optimisation multiplies efficiency of ALL quantum hardware	↑ Moderate	K+ enabling multiplier
ξ	Singapore-centric distribution; limited current geographic reach	→ Neutral	ξ neutral
I	Singapore government champion; ASEAN quantum industrial base	↑ Mild	I+ regional

Risk	Probability	Impact	Mitigation
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Open-source commoditisation	HIGH (35%)	HIGH	Proprietary depth + government moat
Classiq competition	MEDIUM (25%)	HIGH	ASEAN specialisation; Singapore home-ground advantage
Small float volatility	HIGH (55%)	MEDIUM	Position sizing discipline
Pre-revenue dilution risk	MEDIUM (30%)	HIGH	NRF grant coverage; Singapore strategic status

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: SPECULATIVE — SINGAPORE GOVERNMENT MOAT IS THE THESIS

HQ at \$14.50 (mkt cap \$747.9M) is a high-risk compiler software bet anchored by Singapore's quantum national interest. The investment case: compiler IP is needed by ALL hardware companies; Singapore mandate = protected near-term revenue. Key risk: open-source commoditisation. Seldon alignment: K+ (enabling multiplier). HUMAN_ONLY_MANUAL. moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,492ch.